Cubic field automorphisms and Wilson trialities

Human Compression project: L023 working proof

This is an unreviewed replacement for the odd-characteristic generation argument of Leemans–Stokes, arXiv:2208.08215v2, Theorem 3. It does not yet recover that paper's polynomial table or its assertions about the rank-four incidence construction. No complete-original, conjecture, formalization or compression-factor credit is assigned. The source already states the existence result proved below; our contribution is the explicit bound on all obstructing proper subfields.

1 Structural inputs and a subfield count

We retain Dickson's subgroup classification for PSL_2 over a finite field of odd characteristic, in the following precise form. A proper subgroup is contained in a point stabilizer or is cyclic, dihedral, isomorphic to A_4, S_4, A_5, or is a conjugate of a subfield group $\mathrm{PSL}_2(p^d)$ or $\mathrm{PGL}_2(p^d)$. In the last case the defining degree d divides the ambient field degree. The PGL case belongs to the ambient PSL only when the extension degree is even. We also retain the standard automorphism description $\mathrm{Aut}(\mathrm{PSL}_2(p^m)) = \mathrm{PGL}_2(p^m) \rtimes \langle F \rangle$, with F coefficient squaring replaced here by coefficient raising to p. The ambient parameters used below satisfy $p^m \geq 27$, so the small isomorphisms introduce no exception to this description.

The elementary projective-line facts used are these: the involutions of $\mathrm{PSL}_2(r)$ have one $\mathrm{PGL}_2(r)$-conjugacy class; its Klein four subgroups also form one such class and have normalizer S_4 in $\mathrm{PGL}_2(r)$; an involution centralizer in $\mathrm{PGL}_2(r)$ is a dihedral torus normalizer. These familiar consequences of the same classification are retained inputs. The centralizer of a nonidentity element of odd order is cyclic, except for a unipotent element, whose centralizer is an elementary abelian group of odd order. We will only use the resulting fact that such a centralizer contains at most one involution.

Write M_d for a conjugate of the canonical $\mathrm{PGL}_2(p^d)$ inside $\mathrm{PGL}_2(p^m)$, where $d \mid m$.

Lemma 1 (Subfields through a Klein four subgroup)**.** *Fix $V \cong C_2^2$ in $\mathrm{PSL}_2(p^m)$. There are at most four conjugates M_d containing V. If m/d is odd, there is exactly one. In particular, if $m = 3n$, every such M_d with $d \mid n$ lies in the unique conjugate M_n containing V.*

Proof. First $N_{\mathrm{PGL}_2(p^m)}(M_d) = M_d$. The ambient fixed points of the nonidentity unipotent elements of M_d are exactly its projective subline. A normalizer preserves that subline. Use the sharply three-transitive action of M_d on it to make a normalizing projectivity fix three points; it is then the identity. This proves the assertion.

There are two types of Klein four subgroup in $\mathrm{PGL}_2(r)$. Those contained in $\mathrm{PSL}_2(r)$ form the retained single class with normalizer of order 24. For clarity, the other type can be read directly in an involution centralizer. The determinant square-class map on such a subgroup has kernel $\langle z \rangle$ of order two, with $z \in \mathrm{PSL}_2(r)$. In $C_{\mathrm{PGL}_2(r)}(z) = \langle t \rangle \rtimes \langle j \rangle$, the torus has order $r - \varepsilon$, divisible by four, j inverts t, and $z = t^{(r-\varepsilon)/2}$. Both involutions outside $\langle z \rangle$ are reflections. A torus generator has nonsquare determinant, so the reflections outside $\mathrm{PSL}_2(r)$ constitute one of the two reflection classes of this dihedral group. They give one conjugacy class of Klein four subgroups not contained in $\mathrm{PSL}_2(r)$. The normalizer of one has order eight: precisely four torus elements and their four products with one reflection preserve its reflection pair. A normalizer must fix the distinguished element z, so this calculation inside its centralizer is the full normalizer calculation.

Let $X = \mathrm{PGL}_2(p^m)$. Count pairs (M_d, W) with $W \leq M_d$ an X-conjugate of V. For a fixed conjugacy class of Klein four subgroups in M_d, with normalizer order e, the number of conjugates M_d containing V contributed by that class is

$$\frac{|X|}{|M_d|}\frac{|M_d|}{e}\Big/\frac{|X|}{24} = \frac{24}{e}.$$

The class inside its own $\mathrm{PSL}_2(p^d)$ contributes one. The other class either contributes three or contributes nothing, according as it lies in the ambient $\mathrm{PSL}_2(p^m)$ or not. Under an odd extension, determinant square classes are preserved, so it contributes nothing. This gives the stated bounds.

If $d \mid n$, extend any conjugate M_d to the correspondingly conjugated $\mathrm{PGL}_2(p^n)$. It still contains V. Since $m/n = 3$ is odd, this larger group is the unique M_n just proved. □

We use the finite-field form of Hilbert's theorem 90 for matrices: a semilinear map on a finite-dimensional vector space over $\mathbb{F}_{r^3}$ with field part a generator of $\mathrm{Gal}(\mathbb{F}_{r^3}/\mathbb{F}_r)$ and cube the identity is conjugate to the coordinate field map. This is a retained descent theorem, not a new classification premise hidden in the counts. The analogous assertion for projective semilinear transformations follows from it. Indeed the product of a matrix lift with its two twisted copies is a scalar in $\mathbb{F}_r^*$; multiply the lift by a scalar of the inverse norm, which exists because the finite-field norm is surjective. Matrix descent then applies. Consequently the fixed subgroup in $\mathrm{PGL}_2(r^3)$ is a conjugate of $\mathrm{PGL}_2(r)$.

Lemma 2 (Involution counts)**.** *For odd r, the number of involutions in $\mathrm{PGL}_2(r)$ is r^2. The number in $\mathrm{PSL}_2(r)$ is $r(r+\varepsilon)/2$, where $\varepsilon = 1$ for $r \equiv 1 \pmod 4$ and $\varepsilon = -1$ for $r \equiv 3 \pmod 4$.*

Proof. The split and nonsplit involution centralizers in $\mathrm{PGL}_2(r)$ have respective orders $2(r-1)$ and $2(r+1)$. Their class sizes are $r(r+1)/2$ and $r(r-1)/2$, whose sum is r^2. Exactly the first class has square determinant when $r \equiv 1 \pmod 4$, and exactly the second when $r \equiv 3 \pmod 4$. This gives the PSL count as well. □

2 Excluding every proper generated subgroup

Let p be odd, $q = p^n$, $G = \mathrm{PSL}_2(q^3)$ and $X = \mathrm{PGL}_2(q^3)$. Choose a Klein four subgroup $V \leq \mathrm{PSL}_2(p)$ and an element $s \in \mathrm{PSL}_2(p)$ of order three which cycles its three involutions. Such an element is supplied by the A_4 in the normalizer of V, also when $p = 3$. Put

$$\alpha = \mathrm{conj}(s)F^n.$$

The two factors commute and α has order three. Its image in the field-automorphism quotient is nonidentity. Matrix descent shows that $H = C_G(\alpha)$ is a conjugate of $\mathrm{PSL}_2(q)$: an order-three lift of s in $\mathrm{SL}_2(p)$ has norm $s^3 = 1$, and the cubic extension preserves determinant square classes.

Lemma 3. *For any involution $u \in H$, a proper subgroup $J = \langle V, u\rangle < G$ is contained in an α-stable conjugate of some $M_d = \mathrm{PGL}_2(p^d)$ with $d \mid 3n$ and $d < 3n$.*

Proof. The group J is α-stable, and α cycles V while fixing u. It cannot be in a point stabilizer: in odd characteristic the latter's Sylow two-subgroups are cyclic, so cannot contain V. It cannot be cyclic. A dihedral group containing V has an even rotation subgroup, and its central involution is one of the three involutions of V. Unless the whole group is V, that central involution is characteristic and cannot be cycled with the others. If $J = V$, it has no involution fixed by α.

For $J \cong A_4$, an automorphism of order three has no fixed involution, as it cycles the three involutions. For $J \cong S_4$ or A_5, an automorphism of order three is inner, induced by a three-cycle, whose centralizer in J has no involution. These cases also contradict the fixed element

u. The retained subgroup classification therefore makes J an actual subfield group $\mathrm{PSL}_2(p^d)$ or $\mathrm{PGL}_2(p^d)$. The ambient normalizer of the former is its corresponding M_d, and the latter normalizes itself. The same subline argument as in Lemma 1 proves these normalizer statements; for the former, the normalizer contains M_d, which can be used to adjust three points. Since α normalizes J, it normalizes this M_d as well. □

Theorem 4 (Generating fixed involutions)**.** *There is an involution $u \in C_G(\alpha)$ such that $\langle V, u\rangle = G$. More precisely, if $n \geq 2$, the number of such involutions is at least*

$$\frac{q(q+\varepsilon)}{2} - 1 - 4\sum_{v=1}^{\lfloor n/2\rfloor} p^{2v} > 0, \qquad \varepsilon = (-1)^{(q-1)/2}.$$

For $n = 1$ at most one involution of $C_G(\alpha)$ fails to generate together with V.

Proof. Call an involution of H bad if it generates a proper group with V. By Lemma 3, each bad involution lies in a normalized proper M_d containing V.

All degrees $d \mid n$ can be treated together. By Lemma 1, their groups lie in the unique M_n containing V, namely the standard $\mathrm{PGL}_2(q)$. On this group F^n is trivial, so α acts as conjugation by s. Its fixed subgroup contains at most one involution, by the retained odd-element centralizer fact. Thus all these degrees together contribute at most one bad involution, with no repeated charge for nested subfields.

If $d \mid 3n$ but $d \nmid n$, its three-adic valuation is exactly one larger than that of n. Thus

$$d = 3v, \qquad \gcd(d, n) = v, \qquad v \mid n.$$

For a proper degree $d < 3n$ one has $n/v \geq 2$, hence $v \leq n/2$. The restriction of F^n to the degree-d field has order three. On an α-stable conjugate M_d the restriction of α has this same nontrivial field component and order three. Projective matrix descent identifies its fixed subgroup with $\mathrm{PGL}_2(p^v)$, so at most p^{2v} of its elements are involutions. There are at most four such conjugates containing V, by Lemma 1.

Summing over all integers $1 \leq v \leq \lfloor n/2\rfloor$ only enlarges the union of bad candidates. This proves the bound. For $n = 1$ the sum is empty and $|\operatorname{Inv}(H)| \geq 3$. For $n \geq 2$ put $b = \lfloor n/2\rfloor$. Then

$$\sum_{v=1}^{b} p^{2v} < \frac{p^2}{p^2-1} p^{2b} \leq \frac{9}{8} q.$$

Except for $q = 9$, one has $q \geq 25$, and therefore $q(q-1)/2 > 1 + (9/2)q$. For $q = 9$ the exact bound is $45 - 1 - 4 \cdot 9 = 8 > 0$. This proves positivity in every case. □

3 The Wilson class

We retain the usual correspondence between quotients of $\langle r_0, r_1, r_2 \mid r_i^2 = (r_0 r_2)^2 = 1\rangle$ and reflexible maps, together with Wilson's permutation action on $r_0, r_2, r_0 r_2$. This is a map correspondence; no intersection property for a polytopal poset is being inferred from generation.

Corollary 5. *For every odd prime power q, the group $\mathrm{PSL}_2(q^3)$ is the automorphism group of a reflexible map of Wilson class III.*

Proof. Take distinct $\rho_0, \rho_2 \in V \setminus 1$ in the cyclic order of α, and put $\rho_1 = u$ from Theorem 4. They are involutions, the endpoints commute, and they generate G. The automorphism α fixes ρ_1 and cycles the three nonidentity elements of V, so the resulting map has a Wilson triality.

Suppose it also had a Wilson duality. Together with the triality, the corresponding automorphisms would give an S_3 acting on $V \setminus 1$ and fixing u. This action is faithful: an automorphism

fixing V pointwise and u fixes generators of G. But $\mathrm{Out}(G) \cong C_2 \times C_{3n}$ is abelian. The image of the commutator subgroup A_3 of this S_3 must be trivial there, whereas its generator α has nontrivial field component. This contradiction excludes all Wilson dualities. Thus the map belongs to class III. □

Source boundary. The source's Theorem 3 for even q explicitly retains the earlier Jones–Poulton theorem. The proof above supplies all odd q without claiming that every proper α-stable generated subgroup must lie in the degree-n subfield. Degrees $d \nmid n$ are instead counted and excluded collectively. Its scope is map existence and the absence of Wilson dualities; the source's polynomial solution counts, rank-four flag completion, chamber-orbit counts, and full geometric correlation assertions remain separate obligations. The source authors retain attribution for the stated existence theorem. The finite-field descent and subgroup classification are explicitly borrowed; the normalized-subfield union bound is the present replacement argument. No finite enumeration is a premise.